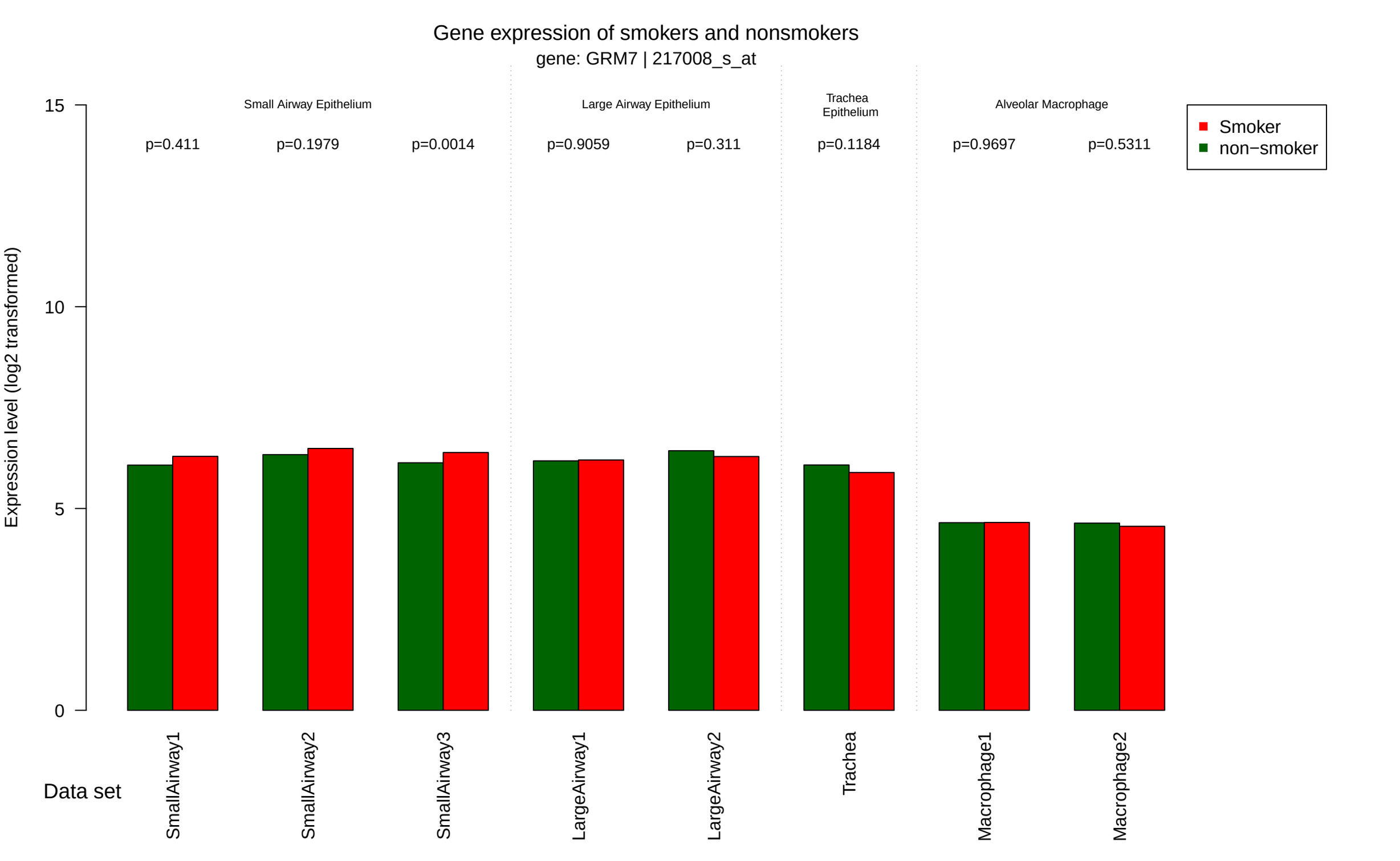
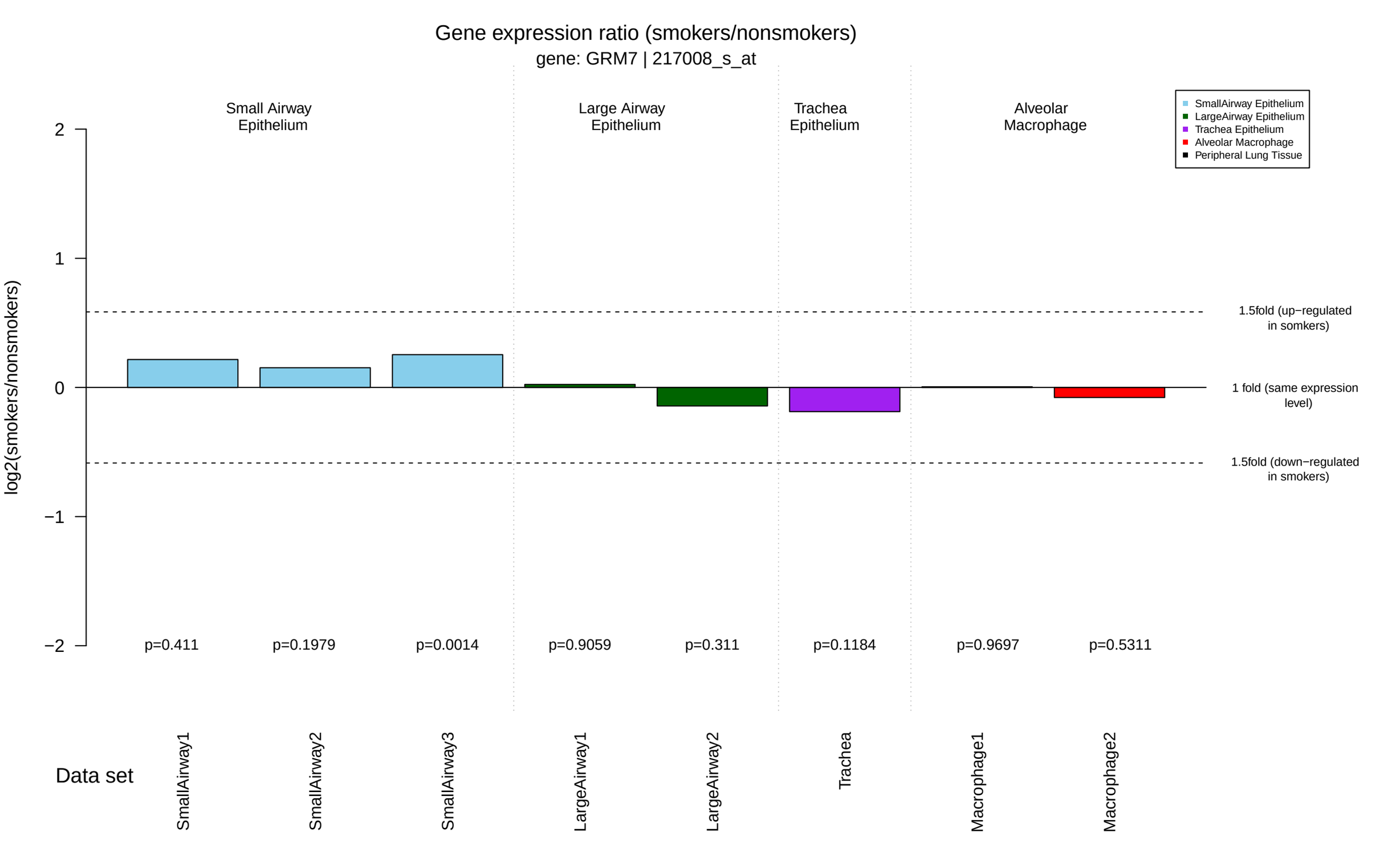
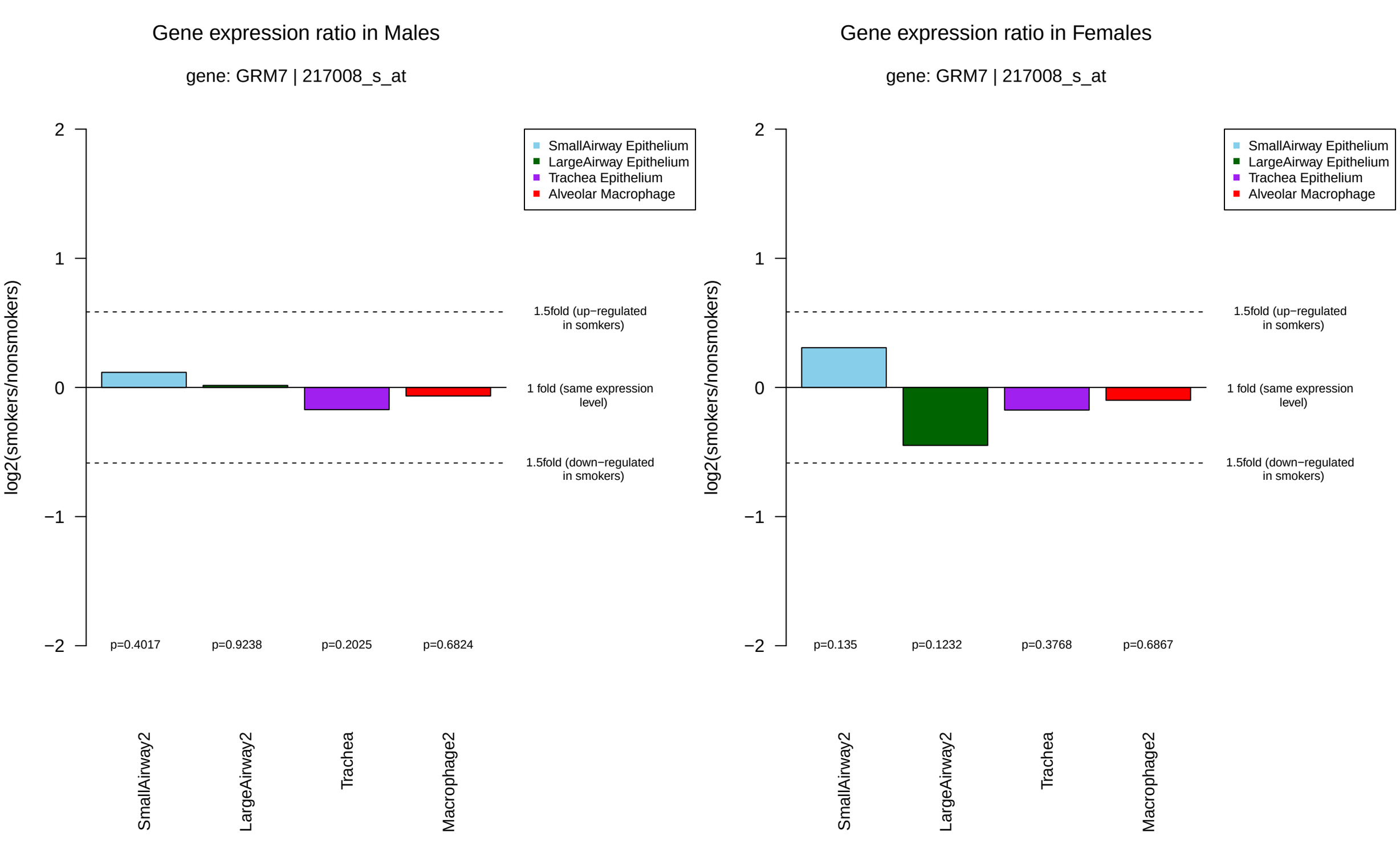
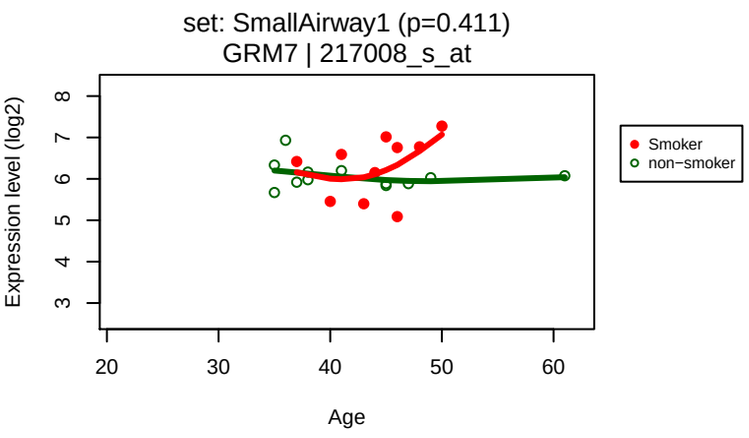
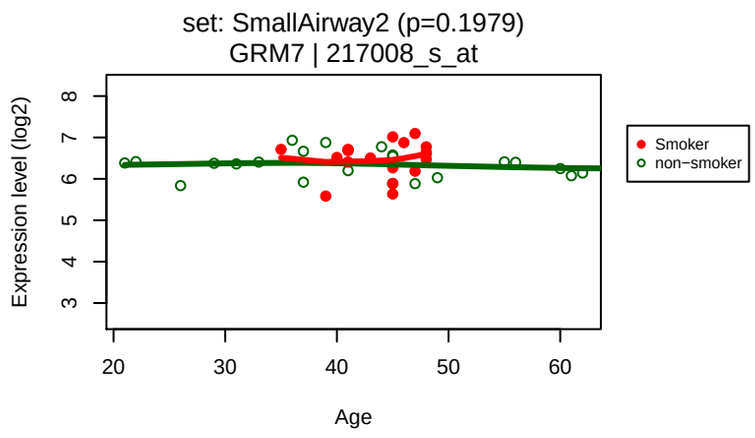
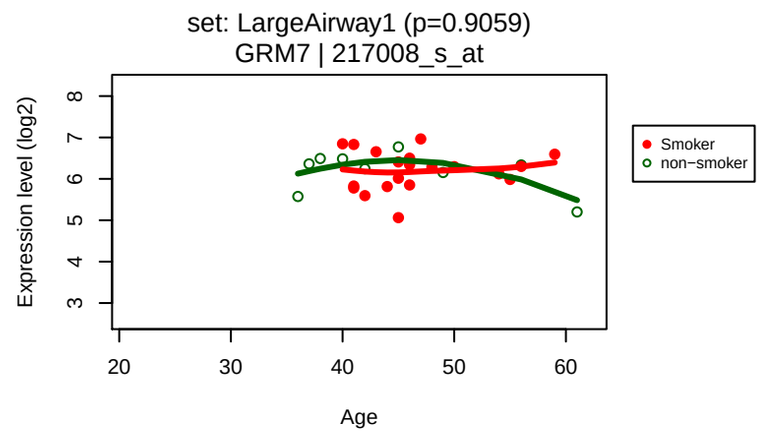

Gene expression of smokers and nonsmokers
gene: GRM7 | 217008_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.411
p=0.1979
p=0.0014
p=0.9059
p=0.311
p=0.1184
p=0.9697
p=0.5311
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

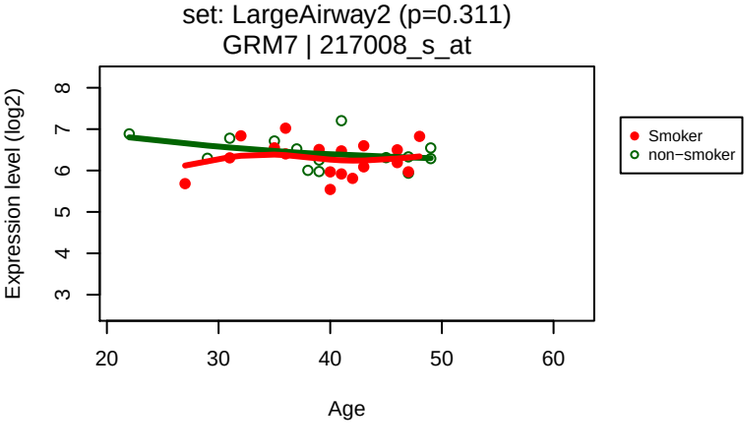

Gene expression ratio (smokers/nonsmokers)
gene: GRM7 | 217008_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.411
p=0.1979
p=0.0014
p=0.9059
p=0.311
p=0.1184
p=0.9697
p=0.5311
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

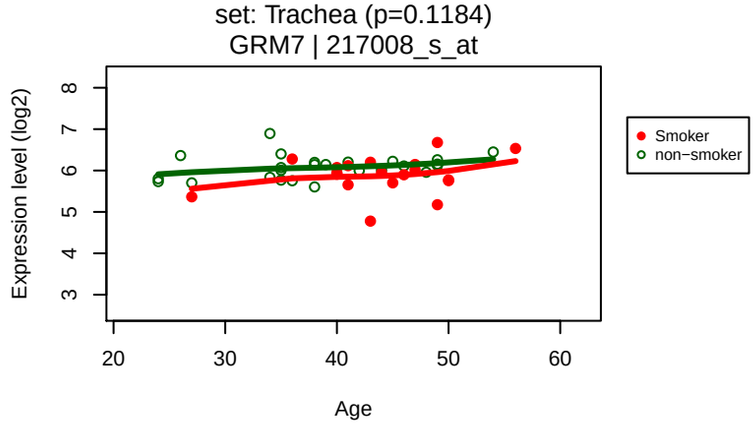

Gene expression ratio in Males
gene: GRM7 | 217008_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4017
p=0.9238
p=0.2025
p=0.6824
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GRM7 | 217008_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.135
p=0.1232
p=0.3768
p=0.6867
SmallAirway2
LargeAirway2
Trachea
Macrophage2

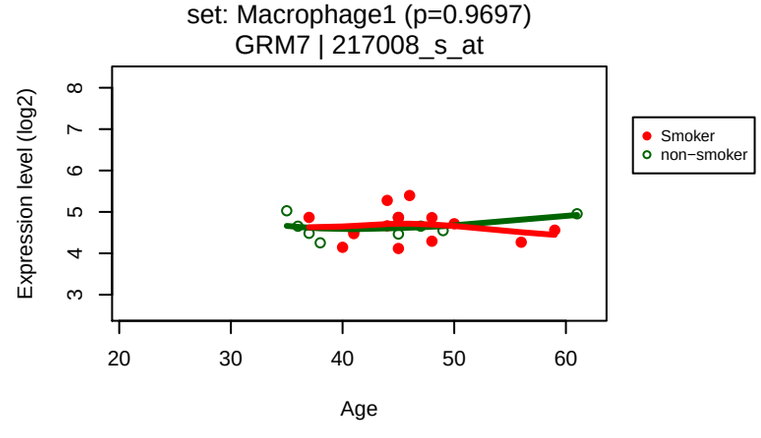

set: SmallAirway1 (p=0.411)
GRM7 | 217008_s_at
Expression level (log2)
Age
Smoker
non−smoker

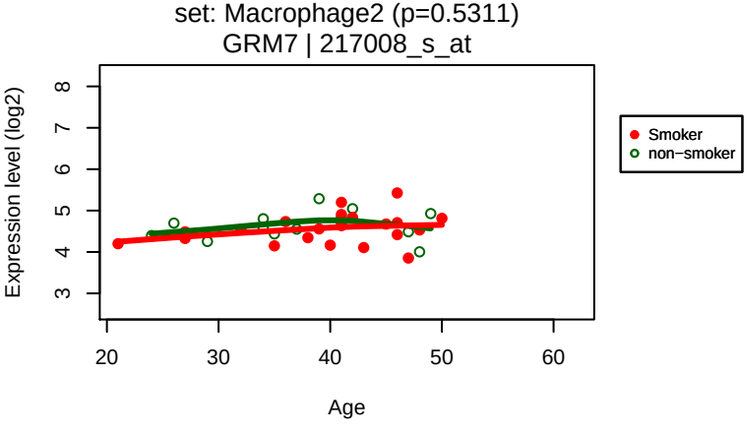

set: SmallAirway2 (p=0.1979)
GRM7 | 217008_s_at
Expression level (log2)
Age
Smoker
non−smoker

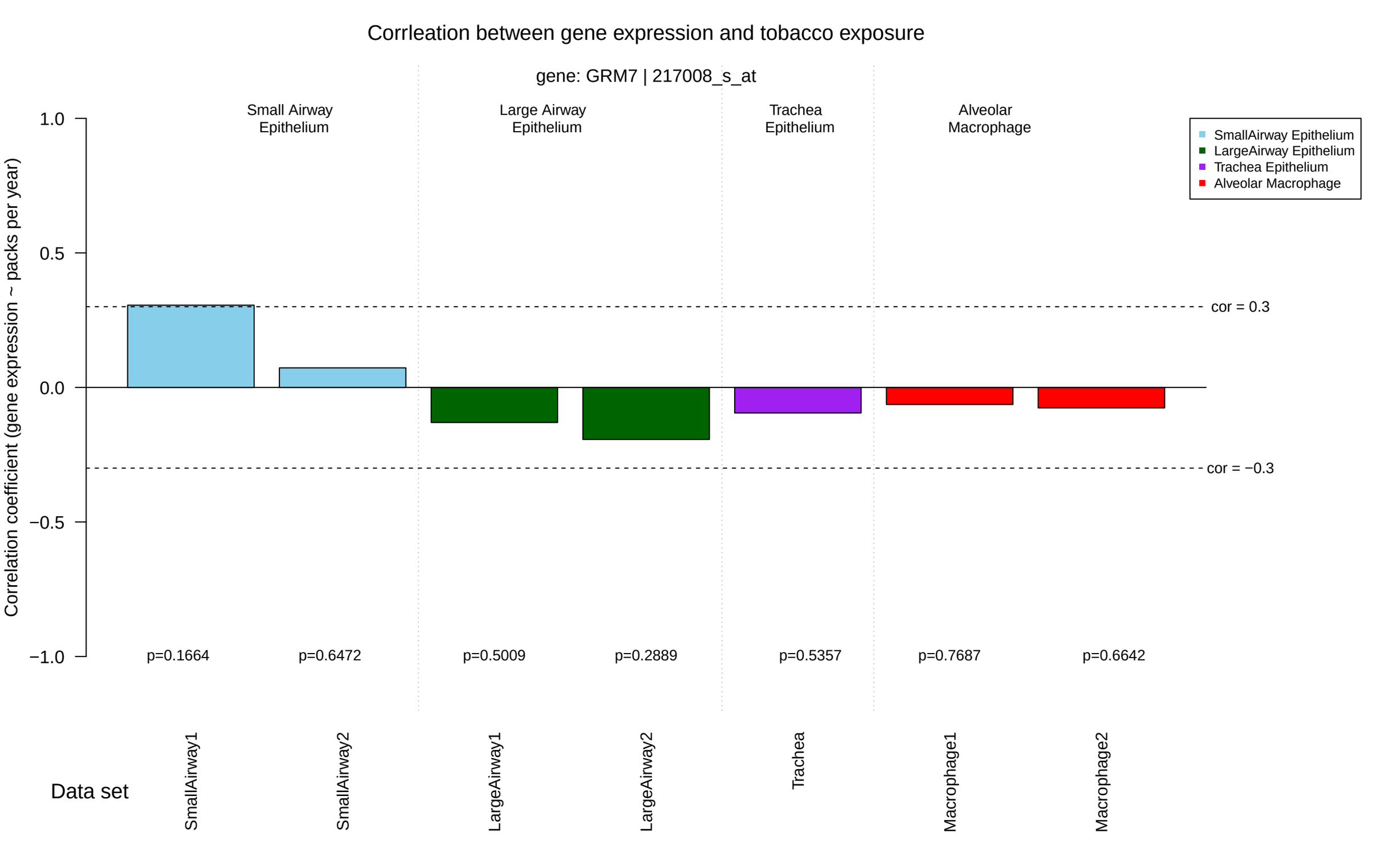

set: LargeAirway1 (p=0.9059)
GRM7 | 217008_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.311)
GRM7 | 217008_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1184)
GRM7 | 217008_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9697)
GRM7 | 217008_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5311)
GRM7 | 217008_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GRM7 | 217008_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1664
p=0.6472
p=0.5009
p=0.2889
p=0.5357
p=0.7687
p=0.6642
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2